1 Gain law

($e_\theta = \theta - \hat{\theta}$; coefficients at the estimate)

$$\bar{a}(\bar{w}, b) = 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)}$$
$$\bar{a}' := \frac{\partial \bar{a}}{\partial \bar{w}} = b\big(1 - \bar{a}\big)^2$$
$$\bar{a}'' := \frac{\partial \bar{a}'}{\partial \bar{w}} = -2\,b^2\big(1 - \bar{a}\big)^3$$
$$A_a := \frac{\partial \bar{a}'}{\partial \bar{a}} = -2\,b\big(1 - \bar{a}\big)$$
$$\Big(\frac{\partial \bar{a}'}{\partial b}\Big)_{\bar{a}} = \big(1 - \bar{a}\big)^2$$
$$\Big(\frac{\partial \bar{a}}{\partial b}\Big)_{\bar{w},\,\bar{w}_s} = \frac{1 - \bar{a}}{b}$$

2 State

($\alpha = 1 - \lambda_c$)

$$\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b,\ x_6,\ x_7,\ m_1,\ m_2\,\big]^\top$$
$$\mathbf{e} = \big[\,e_1,\ e_2,\ e_3,\ e_4,\ e_b,\ e_6,\ e_7,\ e_8,\ e_9\,\big]^\top, \qquad e_j = e_{\delta\bar{w}_j}\ (j \le 3), \quad e_4 = e_{\bar{a}_w}$$

3 Substitution

$$e_{\bar{a}_w}[k] = \psi[k] := -\hat{b}\big(1 - \hat{\bar{a}}_w\big)^2 e_3[k] + \frac{1 - \hat{\bar{a}}_w}{\hat{b}}\, e_b[k]$$

4 Error dynamics

$$F_{12} = F_{23} = F_{55} = F_{66} = F_{77} = F_{98} = 1$$
$$F_{33} = \lambda_c + F_{dw}\,\bar{a}', \qquad F_{34} = 0, \qquad F_{35} = -F_{dw}\,\frac{1 - \hat{\bar{a}}_w}{\hat{b}}, \qquad F_{38} = F_{39} = -\alpha$$
$$F_{43} = (1 - \lambda_c)\,\bar{a}' + J_{d3}M - F_{dw}\,\bar{a}'^2$$
$$F_{44} = 1 + A_a M$$
$$F_{45} = \big(1 - \hat{\bar{a}}_w\big)^2 M + F_{dw}\big(1 - \hat{\bar{a}}_w\big)^3$$
$$F_{48} = F_{49} = \bar{a}'\alpha$$
$$\mathbf{g}_n = \big[\,0,\ 0,\ -(1 - \lambda_c),\ \bar{a}'(1 - \lambda_c),\ 0,\ 0,\ 0,\ 0,\ 0\,\big]^\top$$
$$F_{:,1} \leftarrow F_{:,1} - \mathbf{g}_n$$

5 Measurement

$$H_1 = \begin{bmatrix} 1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0 \end{bmatrix}$$
$$\varepsilon := 1 - S$$

Original

$$H_2 = s\,\varepsilon\Big[\,0,\ 0,\ -\nabla_d \bar{w}_d\,\frac{\partial \bar{a}'}{\partial \delta \hat{\bar{w}}_3},\ 1 - \nabla_d \bar{w}_d\,A_a,\ -\nabla_d \bar{w}_d \big(1 - \hat{\bar{a}}_w\big)^2,\ 0,\ 0,\ 0,\ 0\,\Big]$$

Orange

$$H_{23} = s\,\varepsilon\Big(-\nabla_d \bar{w}_d\,\frac{\partial \bar{a}'}{\partial \delta \hat{\bar{w}}_3} - \big(1 - \nabla_d \bar{w}_d\,A_a\big)\bar{a}'\Big)$$
$$H_{24} = 0$$
$$H_{25} = s\,\varepsilon\Big(-\nabla_d \bar{w}_d \big(1 - \hat{\bar{a}}_w\big)^2 + \big(1 - \nabla_d \bar{w}_d\,A_a\big)\frac{1 - \hat{\bar{a}}_w}{\hat{b}}\Big)$$

With $\partial \bar{a}'/\partial \delta \hat{\bar{w}}_3 = 0$:

$$H_{23} = -s\,\varepsilon\big(1 + 2\hat{b}(1 - \hat{\bar{a}}_w)\,\nabla_d \bar{w}_d\big)\,\hat{b}\big(1 - \hat{\bar{a}}_w\big)^2$$
$$H_{25} = s\,\varepsilon\Big(\frac{1 - \hat{\bar{a}}_w}{\hat{b}} + \nabla_d \bar{w}_d \big(1 - \hat{\bar{a}}_w\big)^2\Big)$$

6 Second-order mean of the gain slot

$$\mathbf{c} = \Big[-(1 - \lambda_c),\ 0,\ (1 - \lambda_c) - F_{dw}\,\bar{a}',\ 0,\ F_{dw}\,\frac{1 - \hat{\bar{a}}_w}{\hat{b}},\ 0,\ 0,\ \alpha,\ \alpha\Big]^\top$$
$$u[k] = \mathbf{c}[k]^\top \mathbf{e}[k] + w_T[k], \qquad \mathrm{Cov}\big(e_j, u\big) = \big(P\,\mathbf{c}\big)_j, \qquad \mathrm{Var}(u) = \mathbf{c}^\top P\,\mathbf{c} + \kappa_T \hat{\bar{a}}_w$$
$$\hat{\bar{a}}_w[k+1] = 1 - \frac{1}{1/\big(1 - \hat{\bar{a}}_w[k]\big) + \hat{b}[k]\,\Delta \hat{\bar{w}}[k]} - \bar{a}''\,\mathrm{Cov}\big(e_3, u\big) - \big(1 - \hat{\bar{a}}_w\big)^2 \mathrm{Cov}\big(e_b, u\big) + \tfrac{1}{2}\,\bar{a}''\,\mathrm{Var}(u)$$

7 Unobservable direction

$$F\,\mathbf{e}_4 = \big(1 + A_a M\big)\mathbf{e}_4$$
$$H_1 \mathbf{e}_4 = H_2 \mathbf{e}_4 = 0$$
$$H\,\Phi(k, k_0)\,\mathbf{e}_4 = 0 \qquad \forall\, k \geq k_0$$

8 Unchanged

$$\hat{y}_2 = s\Big(\hat{\tilde{a}}_w - \varepsilon\, \bar{a}'\, \nabla_d \bar{w}_d\Big)$$

$$Q^{\text{orange}} = Q, \qquad R^{\text{orange}} = R, \qquad P^{\text{orange}}[0] = P[0]$$